

Chinese USSPEE STEM Exams QAE Data

Catalog no.: KS2025M02

(forthcoming, numbers are estimates)

Description:

This dataset offers a comprehensive collection of Question, Answer, Explanation (QAE) triplets drawn from China's University-Specific Exams of the Postgraduate Entrance Exam (USSPEE, 研究生入学考试招生单位自命题科目), the standardized national assessment required for admission to master's degree programs across China.

The Postgraduate Entrance Exam (PEE), commonly known as 考研 (Kaoyan), is a highly competitive exam taken by millions of candidates each year as a prerequisite for graduate study in China.

The PEE consists of two main components. The first is the National Unified Subjects (NUSPEE), designed and administered by the Ministry of Education Examination Center (教育部考试中心). Covering approximately 16 subjects, the NUSPEE serves as China's nationwide academic benchmark, comparable to standardized tests such as the GRE, GMAT, or LSAT.

The second component is the University-Specific Exams (USSPEE), created and administered by individual universities, covering a much broader range of academic disciplines tailored to each institution's requirements.

This dataset covers the STEM portion of the USSPEE exams. NUSPEE exams are provided as a separate dataset, the *Chinese NUSPEE Exams QAE Data* (Catalog No. KS2025M01).

This dataset covers over 100 subject areas. Some of the most popular subjects include:

- Advanced Mathematics
- Mathematical Analysis
- Signals and Systems
- Automatic Control
- Mechanical Design
- Materials Science
- Structural Mechanics
- Computer Science
- Pharmaceuticals
- Circuit Theory
- Data Structures
- Biology

The QAE triplets were extracted from original PDF or MS Word exam files, with each entry including the question, the correct answer, and—where available—a detailed explanation outlining reasoning steps or solution processes. Most of the question-answer pairs also come with explanations, making the dataset particularly valuable for AI training and evaluation.

Data Format:

Each data item is released in two formats: a machine-readable .json (JavaScript Object Notation or JSON) file and a human-readable .md (Markdown) file. Both contain the same underlying content: metadata of the exam and QAE triplets. Mathematical formulas are provided in LaTeX format. Figures are saved as image files, with embedded links included in the JSON and Markdown files.

For convenience, each data item is released in two formats: a machine-readable .json (JavaScript Object Notation or JSON) file and a human-readable .md (Markdown) file. The JSON files are designed for computer processing, making them suitable for automated workflows and data pipelines. The Markdown files present the same information in a more accessible form, ideal for manual inspection and review. They render LaTeX math formulas and display embedded images, providing a clearer, publication-style view of the data.

To view Markdown files in rendered format, you can use common tools such as Visual Studio Code (VS Code) with Markdown preview enabled, or lightweight editors like Typora, which provide real-time rendering of markdown files.

Alternatively, you can open Markdown files in any basic text editor (e.g. Notepad++), though you will not see the rendered formatting.

Language:

Chinese

Data Volume:

10,213 (est.) exams, with a total of 200,000 (est.) QAE triplets.

Data Source:

Chinese universities

Delivery Method:

cloud storage link